📘 **AI/ML Internship – Day 69 Notes & Tasks**

🟢 **Module 9: Vector Databases (ChromaDB & FAISS) – Storing and Searching Embeddings**

📌 Part 1: Recap of Day 68

Yesterday we learned:

✅ Embeddings

✅ Sentence Transformers

✅ Semantic Search

✅ Embedding Generation

Today we will learn:

- What is a Vector Database?
- Why use Vector Databases?
- ChromaDB
- FAISS
- Similarity Search
- Storing and Retrieving Embeddings

📌 Part 2: Why Do We Need a Vector Database?

Suppose a company has:

- 10,000 PDF documents
- 500,000 document chunks
- Millions of embeddings

Can we store them in a Python list?

❌ No.

Searching millions of embeddings manually is extremely slow.

We need a **Vector Database**.

📌 Part 3: What is a Vector Database?

Definition

A **Vector Database** is a specialized database that stores **embeddings (vectors)** and performs **fast similarity searches**.

Instead of searching text directly, it searches vectors based on their meaning.

Simple Definition

A Vector Database is like a **Google Search Engine for embeddings**.

Instead of searching exact words, it searches similar meanings.

📌 Part 4: Traditional Database vs Vector Database

Traditional Database	Vector Database
Stores text, numbers, dates	Stores embeddings (vectors)
Exact matching	Semantic similarity
SQL queries	Similarity search
Best for structured data	Best for AI applications

📌 Part 5: Popular Vector Databases

1. ChromaDB

- Open Source
- Easy to use
- Beginner-friendly
- Excellent for RAG projects

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2. FAISS

- Developed by Meta (Facebook)
- Extremely fast
- Works well with millions of vectors

- Used in large AI systems

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3. Pinecone

- Cloud-based
- Fully managed
- Highly scalable

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4. Weaviate

- Open Source
- AI-native database
- Supports GraphQL

📌 Part 6: Installing ChromaDB

Install:

pip install chromadb

Import:

import chromadb

📌 Part 7: Creating a ChromaDB Collection

```
import chromadb


client = chromadb.Client()


collection = client.create_collection(
    name="college_data"
)
```

A **collection** is like a table in a traditional database.

📌 Part 8: Adding Documents

```
collection.add(
    documents=[
        "AI Internship Fee is ₹5000",
        "Duration is 3 Months",
        "Location is Technopark"
    ],

    ids=[
        "1",
        "2",
        "3"
    ]
)
```

Now the documents are stored inside ChromaDB.

📌 Part 9: Searching Documents

```
results = collection.query(

    query_texts=[
        "How much is the internship fee?"
    ],

    n_results=1

)

print(results)
```

Output:

AI Internship Fee is ₹5000

Notice that the user asked:

"How much is the internship fee?"

The stored document was:

"AI Internship Fee is ₹5000"

Even though the words are different, ChromaDB retrieves the correct result using semantic similarity.

📌 Part 10: Introduction to FAISS

FAISS stands for:

Facebook AI Similarity Search

It is a library developed by Meta for performing **fast similarity searches** on large collections of vectors.

Install:

pip install faiss-cpu

Import:

import faiss

📌 Part 11: FAISS Workflow

Embeddings

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Create FAISS Index

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Store Vectors

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User Question

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Convert to Embedding

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Similarity Search

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Retrieve Best Match

📌 Part 12: ChromaDB vs FAISS

ChromaDB	**FAISS**
Stores documents + metadata	Stores vectors only
Easy to use	Faster for large datasets
Beginner-friendly	Advanced usage
Built for RAG	Built for high-speed similarity search

📌 Part 13: Real-World Applications

Company Knowledge Assistant

Search company policies.

AI Chatbot

Retrieve answers from PDFs.

Recommendation Systems

Recommend similar movies or products.

Search Engines

Semantic document search.

Healthcare

Search patient records and medical documents.

📌 Part 14: Advantages

✅ Very fast searching

✅ Handles millions of vectors

✅ Better semantic search

✅ Essential for RAG

✅ Scalable

📌 Part 15: Limitations

❌ More storage required

❌ Requires embeddings

❌ Incorrect embeddings reduce accuracy

❌ Some vector databases require cloud infrastructure

📌 Part 16: Complete Workflow

PDF

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Extract Text

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Chunking

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Embeddings

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Vector Database

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Similarity Search

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Final Answer

📌 **Part 17: Theory Questions**

What is a Vector Database?

A Vector Database stores embeddings and performs similarity searches based on semantic meaning.

Why do we use ChromaDB?

ChromaDB stores document embeddings and allows efficient semantic retrieval for RAG applications.

What is FAISS?

FAISS is an open-source library developed by Meta for high-speed similarity search over vector embeddings.

What is a Collection in ChromaDB?

A Collection is a container used to store documents, embeddings, and metadata.

Why is a Vector Database important in RAG?

It retrieves the most relevant document chunks efficiently, enabling accurate AI-generated answers.

📌 Part 18: Interview Questions

What is the difference between ChromaDB and FAISS?

ChromaDB stores documents, metadata, and embeddings, while FAISS focuses on high-performance vector similarity search.

Why can't we use a normal database for RAG?

Traditional databases perform exact matching, whereas RAG requires semantic similarity search using embeddings.

What is Similarity Search?

It is the process of finding the vectors closest in meaning to a query vector.

Name three vector databases.

- ChromaDB
- FAISS
- Pinecone

Which vector database is recommended for beginners?

ChromaDB, because it is simple to use and integrates well with RAG projects.

🎯 Tasks for Day 69

🧠 Theory

1. Define a Vector Database.
2. Explain ChromaDB.
3. Explain FAISS.
4. Differentiate ChromaDB and FAISS.
5. Why are vector databases used in RAG?

💻 Practical Task 1

Install:

pip install chromadb

Create a collection named **college_data**.

💻 Practical Task 2

Insert five sample documents into the collection.

Example:

- AI Fee
- Duration
- Eligibility
- Hostel
- Placement

💻 Practical Task 3

Search for:

How long is the internship?

Observe the retrieved result.

💻 **Practical Task 4**

Install FAISS:

pip install faiss-cpu

Read about how FAISS indexes vectors.

💻 **Practical Task 5**

Prepare a comparison table of:

- ChromaDB
- FAISS
- Pinecone

Compare based on:

- Open Source
- Cloud Support
- Ease of Use
- Best Use Cases